MEDICINE MOLECULAR BIOLOGY

CRISPR-Cas9 Gene Editing

ExaminAI Academic Suite | quick-summary | 12/9/2026

1. Overview of CRISPR

CRISPR-Cas9 provides precise targeted genome modification. Originally discovered as an adaptive immune mechanism in bacteria against bacteriophages.

KEY

The single guide RNA (sgRNA) pairs with genomic DNA upstream of a Protospacer Adjacent Motif (PAM).

Mechanism of Action

1. Recognition of the 5'-NGG PAM sequence by Cas9 endonuclease.
2. Unwinding of DNA duplex and RNA-DNA hybrid formation.
3. Generation of double-strand break (DSB) 3 base pairs upstream of PAM.

- [x] sgRNA synthesis
- [x] PAM target verification
- [ ] Cellular delivery via lentivirus